KodBTW

Unified Coding Identity & Campus Coding Intelligence Platform

Project Based Learning – Advanced Java Laboratory (Java Projects), 5th Semester
Project code PBL2627-CSE-C-064 | Session 2026-27
Guide: Er. Ram Babu Buri, Department of CSE

Sajal Goyal · Shri Krishna Chandra Vaishnav · Yug Mittal · Ravi Shah · Suryadev Singh

Coding identity is scattered across platforms

Fragmented

Each platform has its own profile, so users switch between services to review their progress.

No history

Platform pages show the current state; a timeline has to be recorded by hand.

Not comparable

Ratings and “solved” counts are defined differently on each platform.

A unified view is only useful if it shows source, observation time and what is missing.

What KodBTW provides



Accounts and profiles

Registration, login, email verification, Google flows, unique public username.



Handles and sync

Owner-scoped platform accounts; sync only for enabled sources.



Dated snapshots

Source-labelled observations power history and analytics.



Dashboard

Metrics, platform share, difficulty, history, activity and insights.



Leaderboard and contests

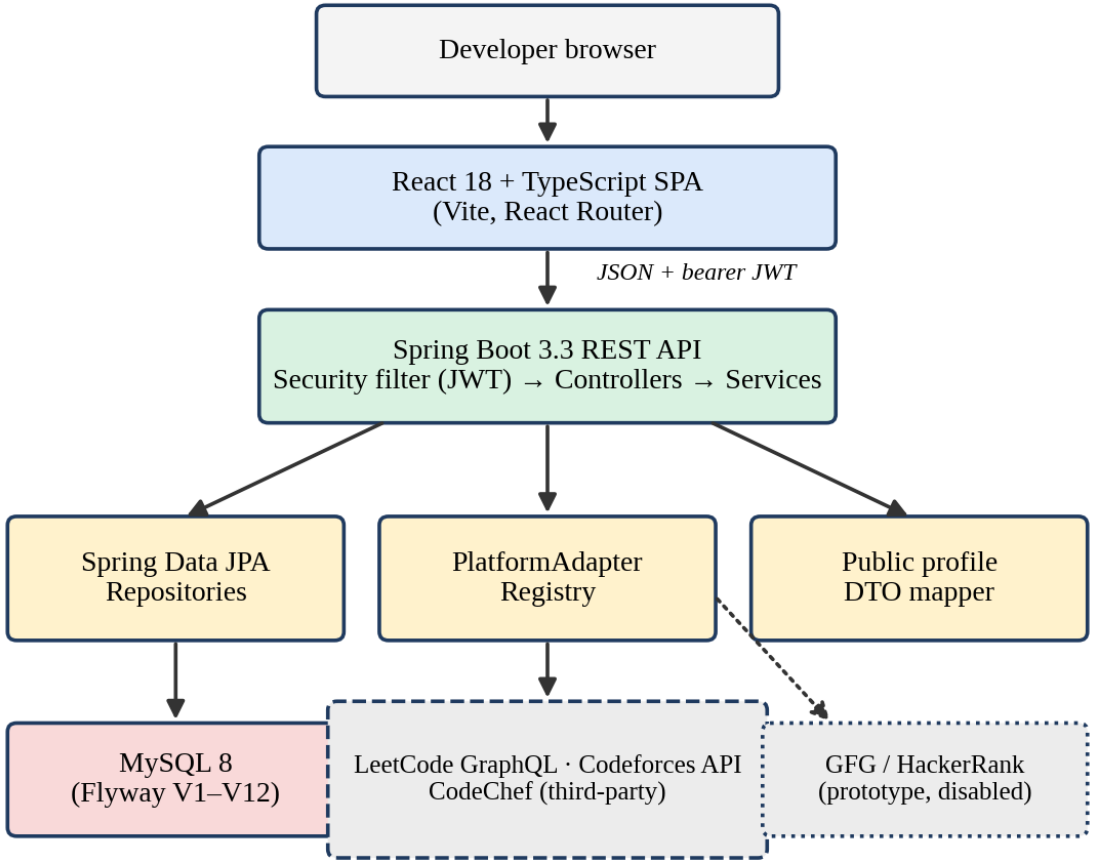
Cached, paginated leaderboard; Codeforces contest calendar.



Public profile

Sanitised DTO at /u/:username; indexing is opt-in.

Layered, stateless client–server architecture



- React 18 + TypeScript SPA sends JSON to the Spring Boot REST API
- Protected calls carry a bearer JWT; a security filter resolves the user
- Controllers delegate to services for ownership checks and logic
- Services use JPA/MySQL or a platform adapter
- Responses are explicit DTOs

Technology stack

Java 21

Spring Boot 3.3.5

React 18

TypeScript 5.6, Vite 5.4

MySQL 8

Flyway migrations V1–V12

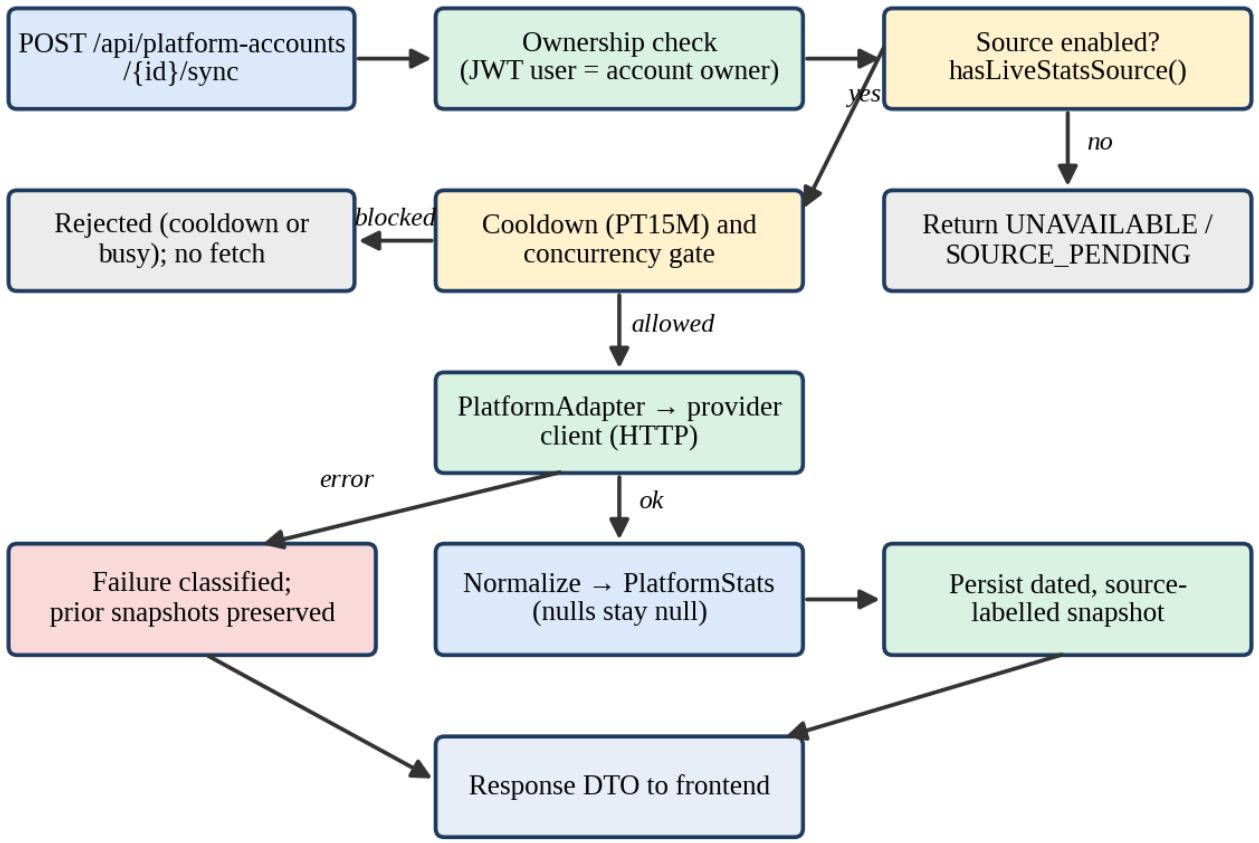
Layer	Technologies
Frontend	React 18, TypeScript 5.6, React Router 6, Vite 5.4
API	Java 21, Spring Boot 3.3.5, Spring Web, validation
Security	Spring Security, JWT (JJWT), BCrypt password encoder
Persistence	Spring Data JPA, Hibernate, MySQL 8
Schema	Flyway; Hibernate ddl-auto: validate
Backend tests	JUnit, Spring Boot Test, H2, Security Test, Testcontainers
Frontend tests	Node.js built-in test runner

Platform integration: 3 of 5 enabled

Platform	Normal sync	Source label	Key caveat
LeetCode	**Enabled**	LEETCODE_REAL	Undocumented public GraphQL query; terms and stability need review
Codeforces	**Enabled**	CODEFORCES_REAL	Public user.info; solve counts/history may stay null
CodeChef	**Enabled**	CODECHEF_THIRD_PARTY	Unofficial third-party provider; permission and rights unresolved
GeeksforGeeks	**Disabled**	Unavailable / pending	Prototype files fail the live-source gate
HackerRank	**Disabled**	Unavailable / pending	Prototype files fail the live-source gate

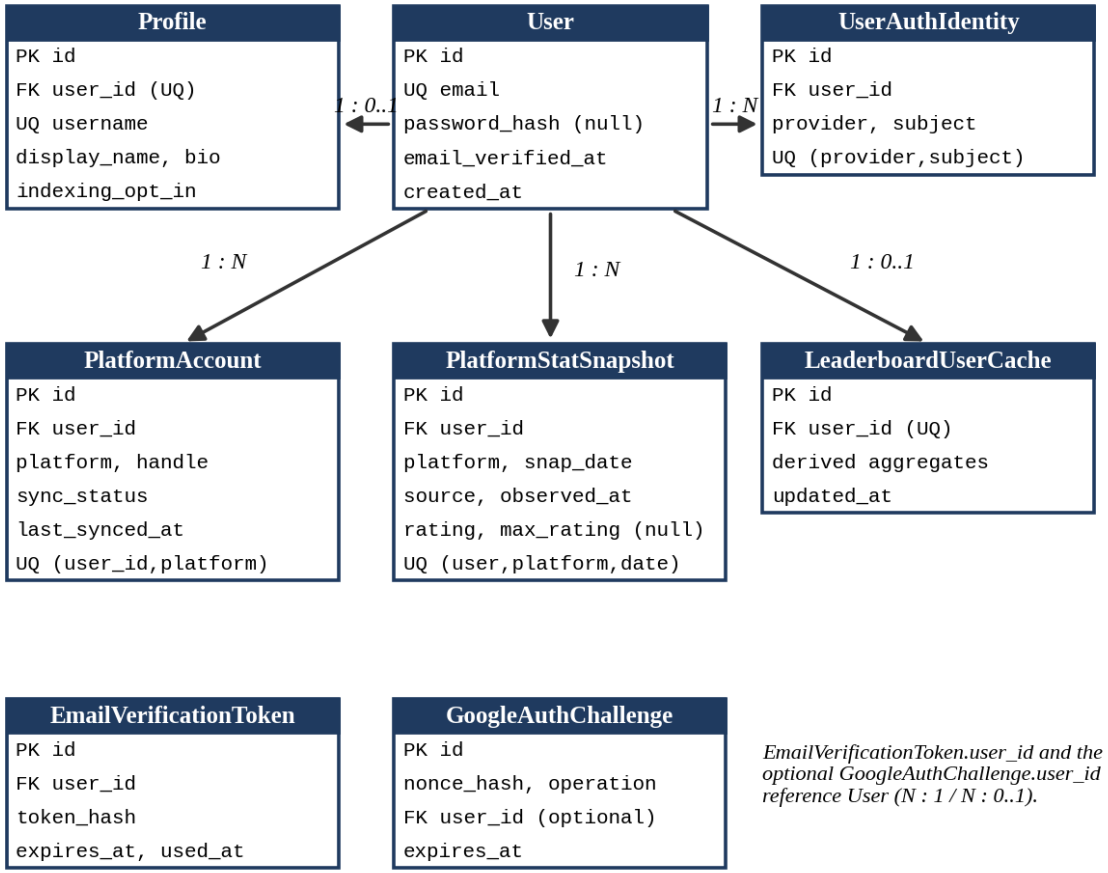
A public page that loads is not authorisation to extract data. No provider is presented as cleared for production.

Sync keeps provenance and never fabricates



- Ownership is checked on every account operation
- Cooldown (default PT15M) and concurrency gates limit provider calls
- Adapters map only fields the provider supplies
- Null means “no supported value”, never zero
- A failed sync keeps earlier snapshots; no mock data replaces them

Data model: dated, source-labelled snapshots



- One profile per user; unique public username
- Unique (user, platform) account; unique (user, platform, date) snapshot
- Max rating and rank stored separately
- Federated identity, Google challenge and email-token tables
- Flyway V1–V12; the migrations are the authoritative DDL

Security, privacy and responsible acquisition



Stateless JWT

Form login and HTTP Basic disabled; secrets supplied at runtime, never committed.



Owner-scoped access

Account operations are checked against the authenticated owner; leaderboard rebuild is ADMIN-only.



Sanitised public profile

Dedicated DTO instead of the raw user entity; search indexing is opt-in.



Source discipline

Disabled sources stay off; third-party data is labelled as such.

Connected ≠ verified

Adding a handle does not prove the user owns it. A displayed handle means “supplied by the user” until a verification flow exists.

Provider terms, data rights and retention still need review before production.

Testing: two historical records, not re-run

Record A

Supplied previously by the project owner

309	backend tests: 0 failures, 0 errors, 2 skipped
110	frontend tests passed; build passed
V1–V11	Docker-backed MySQL migrations passed

Record B

Repository validation doc, run of 12 Oct 2026

262	backend enumerated: 1 Docker error, 2 skipped
261	non-Docker run: 0 failures, 0 errors
70	frontend tests passed; build passed

The records differ and cannot be reconciled without logs and commit IDs. Record A does not cover V12. Re-run on a clean checkout before submission.

Limitations and future work

Limitations

- Upstream schemas, terms and availability can change
- Only 3 of 5 platforms sync; ratings are not equivalent across platforms
- No user study, performance data or production deployment evidence
- No genuine runtime screenshots yet

Future work

- Official or approved data contracts; enable GFG/HackerRank only once cleared
- External-handle ownership verification
- Frontend CI, end-to-end tests, readiness endpoint
- Evaluate educational benefit through a designed study

Conclusion

One authenticated, owner-scoped view of coding records – with source, time and missingness kept visible.

- React/TypeScript client + Spring Boot REST API + MySQL (Flyway V1–V12)
- Adapter-driven sync for LeetCode, Codeforces, CodeChef; dated snapshots
- Dashboards, leaderboard, contest calendar, sanitised public profiles
- No learning benefit or production deployment is claimed